

GarmentERP BD

Complete Enterprise Resource Planning & Quality Assurance System Architecture Specification

GarmentERP BD is a mission-critical, enterprise-grade ERP and Quality Management System engineered specifically around the authentic operational workflows of the Bangladesh Ready-Made Garments (RMG) manufacturing sector. It unites the complete factory journey from **Buyer Inquiries & POs** to **Costing, BOM, Procurement, Warehouse Inventory, Production (Cutting, Sewing, Finishing), Carton Ratio Packing, Comprehensive QA/QC (Fabric 4-Point, Inline DHU, Rework, CAPA, AQL 2.5)**, and **Commercial Shipment**.

31

INTEGRATED WORKFLOWS

14

ROLE-BASED PERSONAS

8

DEVELOPMENT PHASES (100% COMPLETE)

96 / 96

AUTOMATED SQA TESTS PASSED

8

POSTMAN SQA COLLECTIONS

100%

TRACEABLE PO PIPELINE

Executive Architecture Principle: Interconnected Single Source of Truth

GarmentERP BD is engineered as an interconnected business ecosystem. Quantities and specifications are logically coupled across departments—a quantity adjustment in a Buyer PO automatically cascades into BOM requirements, triggers Procurement shortage alerts, governs Production cut plies, enforces Finishing carton limits, gates Quality thresholds, and controls Export commercial invoices.

Architect & Author

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Target Industry

Bangladesh RMG Garment Manufacturing (Savar, Gazipur, Chattogram)

Document Version & Date

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1. Executive Summary & System Architecture

The Ready-Made Garments industry in Bangladesh represents over 84% of national exports. However, modern manufacturing plants often suffer from operational silos, where Merchandising, Commercial, Stores, Production, and Quality teams operate on disparate spreadsheets, paper tickets, and disconnected software. GarmentERP BD replaces this fragmentation with an auditable, real-time platform.

1.1 Architectural Topology



1.2 Core Technology Stack Summary

TIER	TECHNOLOGY	SPECIFICATION & ARCHITECTURAL PURPOSE
Frontend	React 18, TypeScript, Vite, Tailwind CSS	Zero-lag Single Page Application built on modern enterprise light theme design guidelines (#F7F8FA background, #FFFFFF cards, Deep Green #047857 primary accent).
State & Forms	TanStack Query, React Hook Form, Zod	Optimistic server-state caching, strict client-side validation schemas, and real-time query invalidation.
Backend API	Node.js, Express.js, TypeScript	Type-safe RESTful API architecture with strict modular domain separation, centralized middleware, and error handling.
Security	JWT, bcryptjs, RBAC Middleware	Stateless access tokens with refresh tokens, password hashing, and segregation of duties across 14 personas.
Database	PostgreSQL & Prisma ORM	Relational schema modeling Bangladesh factories, buildings, floors, lines, and warehouses.
Quality Assurance	Node Test Runner, Postman, Newman	Automated regression test suites covering positive, negative, and boundary conditions across all ERP endpoints.

1.3 UI/UX Design System Guidelines

In accordance with enterprise ERP standards, GarmentERP BD strictly avoids gimmicky dark dashboards, neon accents, or bloated admin templates. It adheres to an **Executive Light Theme**:

Design Tokens & Palette	Enterprise Principles
Background Canvas #F7F8FA (Very Light Gray)	Collapsible Navigation Thin right border, grouped modules
Surfaces & Cards #FFFFFF (Pure White)	Compact KPI Cards Large numbers with subtle percentage indicators
Primary Brand #047857 / #065F46 (Deep Green)	Data-Dense Tables Search, filter, badges, and export actions
Secondary / Badges #ECFDF5 (Soft Tint Green)	Order Details Tabs Comprehensive 360° progress pipeline
Alerts / Warnings #F59E0B (Amber / Gold)	Role Switcher 1-click instant persona switching for testing
Text & Typography #0F172A (Dark Slate / Inter)	
Borders #E2E8F0 (Subtle Slate Border)	

2. 14 Role-Based Access Personas & Factory Topology

Strict segregation of duties is mandatory under international social compliance standards (WRAP, BSCI, SEDEX). GarmentERP BD enforces role-based access control across 14 factory roles:

2.1 The 14 Specialized Personas

ROLE PERSONA	CODE	FACTORY SCOPE & SEGREGATED OPERATIONAL RESPONSIBILITIES
Super Administrator	SUPER_ADMIN	Complete platform sovereignty, user provisioning, global permissions, and audit log administration.
Factory Administrator	FACTORY_ADMIN	Factory layout management (Savar & Gazipur units), floor and line setup, and staff roster assignments.
Executive Management	MANAGEMENT	Overall operational oversight, order profitability, costing sheet sign-offs, and shipment authorizations.
Merchandiser	MERCHANDISER	Buyer relationships, style masters, tech pack versioning, consumption BOMs, and order tracking.
Purchase Officer	PURCHASE_OFFICER	Purchase requisitions, supplier scorecards, supplier PO issuance, and raw material procurement.
Store Officer	STORE_OFFICER	Multi-warehouse stock control (Fabric, Trims, Accessories, FG), GRN entry, negative stock prevention.
Production Planner	PROD_PLANNER	Line capacity allocation, target SMV calculations, cutting schedules, and sewing plan balancing.
Production Supervisor	PROD_SUPERVISOR	Hourly sewing floor tracking (Lines 01-04), cut bundle intake, machine downtime, and output verification.
QA Manager	QA_MANAGER	Quality standards enforcement, CAPA 5-Whys verification, AQL audit approvals, and shipment quality gates.
QC Inspector	QC_INSPECTOR	4-Point fabric inspection, cutting panel QC, inline sewing defect logging (DHU), and rework inspection.
Commercial Officer	COMMERCIAL_OFFICER	Export documentation, Letters of Credit (LC), customs invoices, packing lists, and security gate passes.
HR Officer	HR_OFFICER	Employee attendance, shift rosters, overtime computation, biometric integration, and monthly payroll.
Finance Officer	FINANCE_OFFICER	Cost center allocations, supplier payables, buyer receivables, CM earnings, and order net margin analysis.
Maintenance Officer	MAINTENANCE_OFFICER	Sewing machine maintenance schedules, breakdown work orders, spare parts consumption, and downtime.

2.2 Factory Topology & Spatial Hierarchy (Module 02)

Enterprise Holding (Apex Garments Holdings Ltd.)

- Plant 01: Savar Manufacturing Complex (Code: AG-SAVAR-01)
 - Building 01 (Main Production Block)
 - Floor 01: Warehouse & Fabric Store (Bins: FAB-01 to FAB-18)
 - Floor 02: Automated Spreading & Cutting Floor (Tables C1 - C6, 24h Relaxation Zone)
 - Floor 03: Sewing Department (Production Lines 01 to 04 with Juki DDL-9000C)
 - Floor 04: Finishing, Metal Detection & Export Packing Section
 - Building 02 (Commercial & Admin Center)
- Plant 02: Gazipur Washing & Denim Unit (Code: AG-GAZI-02)
 - Floor 01: Wet Processing & Sustainable Washing Department
 - Floor 02: Finishing, AQL Final Audit Bay & Finished Goods Bonded Store

3. Commercial & Supply Chain Modules (Phases 2 & 3)

The commercial and supply chain backbone ensures that every yard of fabric and button ordered matches the exact requirements of international buyer contracts, preventing stock hoarding and negative inventory.

3.1 Commercial Workflow (Buyer PO, Tech Pack & Costing)

Buyer & Tech Pack Engine (Mod 04 & 05)

Supports global buyers (e.g., H&M, Zara, PVH, XYZ Fashion) with multi-currency (USD, EUR, GBP, BDT) and international payment terms (At-Sight LC, Usance LC). Tech packs feature strict **version control (v1, v2, v3)**, ensuring that revision changes in measurement charts or tolerances (+/- 0.5cm) notify production supervisors immediately.

BOM & Pre-Costing Engine (Mod 07 & 08)

Calculates comprehensive garment unit costs: Fabric, Trims, CM (Cost of Making), Wash, Print/Embroidery, Commercial Overhead, and Net Profit Margin. The BOM explosion multiplies item consumption (e.g. 0.85 KG single jersey/piece) by PO volume to compute gross requirements and auto-detect raw material shortages.

3.2 Supply Chain & Inventory Control (Mod 09 & 10)

Raw materials represent 60–70% of RMG order value. GarmentERP BD automates procurement from shortage detection to warehouse bin allocation:



3.3 Strict Negative Inventory Protection Engine

Core Guard: Hard Negative Stock Prevention

The system strictly disallows stock balance deductions that would result in negative quantities. If a supervisor attempts to issue 8,000 meters from a fabric lot containing only 5,000 meters, the API transaction is immediately rejected with **400 Bad Request: Insufficient Stock Available** and an audit violation is logged. Over-issuance is strictly locked unless an authorized Factory Administrator enters a certified override code.

TRANSACTION TYPE	TRIGGER EVENT	SOURCE LOCATION	DESTINATION	VALIDATION RULES
GRN Receive	Vendor delivery at security gate	Supplier Inbound Truck	Quarantine Inspection Bay	Matches SPO quantity (+/- 5% allowance) & QA certificate.
QC Bin Transfer	Fabric 4-point inspection pass	Quarantine Bay	Bonded Fabric Warehouse	Roll tagged with barcode, length, and dye lot number.
Line Issue	Daily production cutting plan	Fabric Warehouse	Cutting Floor Lay Table	Deduction must be ≤ Available Unreserved Stock.
Stock Adjustment	Monthly physical inventory count	Warehouse Bin	Adjustment Ledger	Requires dual-signature from Store Officer and Finance.

4. Garments Production Management (Phase 4)

Production execution requires synchronized coordination across Cutting, Sewing, Finishing, and Carton Packing. GarmentERP BD provides end-to-end floor visibility and automated bundle routing.

4.1 Cutting Workflow & 24-Hour Fabric Relaxation Timer (Mod 12)

In knitted garment manufacturing, elastane/cotton fabric rolls undergo high mechanical tension during knitting and dyeing. Spreading rolls immediately causes post-cut shrinkage and dimensional failures.

24h Relaxation Countdown Engine

Upon receiving fabric from the warehouse, the roll enters the unrolling cradle and a **24-Hour Digital Relaxation Timer** is initiated. The system locks cut order generation against that roll until the tension-release countdown reaches zero and QC certifies dimensional stability.

Marker Efficiency & Spreading Lay

Tracks marker length, fabric width, plies count, and marker efficiency percentage (typically 84–89%). Minimizes end-loss wastage and generates automated piece yield summaries for fabric reconciliation against BOM standards.

4.2 QR / Barcode Bundle Ticketing Registry

Once cutting is executed, cut panels are bundled into standard lots (e.g., 250 pieces per bundle). Each bundle receives a tamper-evident QR code ticket encoding: Cut Order ID, Bundle #, Size, Color, and Piece Serial Range (e.g., #001 to #250). Bundles are scanned as they move across workstations for real-time WIP (Work-in-Progress) tracking.

4.3 Sewing Line Hourly Tracking (Mod 13)

Tracks production floor output hour-by-hour across lines (e.g., Lines 01 to 04 in Savar Unit) against target Standard Minute Values (SMV):

Line ID	Assigned Garment	Hourly Target	Actual Made	Rejects/Rwk	Line Efficiency
Line 01	Mens Polo Shirt	150 pcs/hr	142 pcs/hr	3 / 2 pcs	94.6% (On Target)
Line 02	Basic Crewneck	180 pcs/hr	175 pcs/hr	4 / 3 pcs	97.2% (Optimal)
Line 03	Raglan Hoodie	120 pcs/hr	102 pcs/hr	6 / 5 pcs	85.0% (Warning)
Line 04	Chino Trouser	110 pcs/hr	105 pcs/hr	2 / 1 pcs	95.4% (Optimal)

4.4 Finishing Section & Ratio Carton Packing (Mod 14 & 15)

Stitched garments move through 4 finishing stages: **Thread Trimming & Suction** → **Steam Pressing & Ironing** → **Folding & Hangtag Attachment** → **Calibrated Needle/Metal Detection**.

Packing Ratio Assortment & Over-Packing Security Guard

GarmentERP BD enforces buyer-specified carton ratio matrices (e.g., Carton #001: Size S:10, M:20, L:20, XL:10 = 60 pieces). Furthermore, the packing module strictly prevents Packed Quantity > Available Finished Quantity, guaranteeing zero phantom carton records.

5. Core QA/QC Suite — The Quality Flagship (Phase 5)

Quality assurance is the crown jewel of GarmentERP BD. It embeds international inspection methodologies across raw materials, in-process sewing, rework loops, root-cause CAPA, and pre-shipment AQL auditing.

5.1 Fabric 4-Point Inspection Engine (ASTM D5430 / Mod 16)

Every incoming fabric roll undergoes 4-point penalty inspection. Defects are assigned points based on physical defect size:

DEFECT LENGTH / SIZE RANGE	PENALTY POINTS	INDUSTRY STANDARD FORMULA & ACCEPTANCE THRESHOLD
Defects up to 3 inches	1 Point	$\frac{\text{Points per } 100 \text{ Sq. Yards}}{\text{Yards}} = \frac{\text{Total Penalty Points}}{\text{Inspected Yards}} \times \text{Fabric Width (Inches)}$ FIRST QUALITY PASS: ≤ 28.0 POINTS REJECTED / SECOND QUALITY: > 28.0 POINTS
Defects over 3 inches up to 6 inches	2 Points	
Defects over 6 inches up to 9 inches	3 Points	
Defects over 9 inches or any hole	4 Points	

5.2 Inline Sewing Quality & Real-Time DHU Engine (Mod 17)

QC roving inspectors record defects at operator stations in real-time. Defect categories include Broken Stitch, Skip Stitch, Open Seam, Puckering, Shading, and Oil Stains. The system automatically computes:

$$\text{DHU (Defects per Hundred Units) \%} = (\text{Total Defect Count} / \text{Total Inspected Pieces}) * 100$$

5.3 Closed-Loop Rework Quarantine & Re-Inspection (Mod 18)



5.4 Root-Cause CAPA 5-Whys Analysis (Mod 19)

When high-severity defect patterns emerge, QA Managers launch Corrective and Preventive Actions (CAPA) utilizing the standardized 5-Whys Root-Cause Methodology:

Case Study: Excessive Skip Stitches on Sewing Line 02

1. Why did the defect happen?	Needle thread skipped bobbin hook during placket stitch.
2. Why did it skip the hook?	Needle temperature was too high, causing thermal thread deflection.
3. Why was temperature high?	Wrong needle point geometry (Standard R used instead of SES ballpoint).
4. Why was wrong needle used?	Mechanic ran out of SES needles and fitted standard needle.
5. Root Cause (5th Why):	Preventive maintenance store had no automatic re-order point for SES needles.

Corrective Action: Fitted titanium SES ballpoint needle. | Preventive Action: Established minimum safety stock in ERP with automated purchase triggers. | Verified by: QA Manager.

5.5 ISO 2859-1 / ANSI ASQ Z1.4 Normal Level II AQL Sampling (Mod 20)

Pre-shipment final audits calculate sampling batches dynamically based on production lot sizes:

ORDER LOT SIZE RANGE	INSPECTION LEVEL	SAMPLE SIZE	CRITICAL (AQL 0)	MAJOR (AQL 2.5)	MINOR (AQL 4.0)
3,201 to 10,000 pcs (Code K)	Level II Normal	200 pieces	Ac: 0 Re: 1	Ac: 10 Re: 11	Ac: 14 Re: 15
10,001 to 35,000 pcs (Code L)	Level II Normal	315 pieces	Ac: 0 Re: 1	Ac: 14 Re: 15	Ac: 21 Re: 22

6. Phase 6: Commercial Invoicing, Export Packing & Security Gate Pass Suite

Phase 6 governs the international trade, commercial documentation, and physical factory gate dispatch of certified export garments, bridging production floor completion with Chittagong Port export logistics.

6.1 Pre-Shipment ISO 2859-1 Quality Gate Security Lock (Mod 21)

Strict Quality Gate Security: Unauthorized Shipment Blocked

Under international buyer shipping protocols, GarmentERP BD enforces an uncompromising system-level lock: Any attempt to issue a Commercial Invoice, Export Packing List, or Security Gate Pass for a Buyer PO that lacks an authorized ISO 2859-1 ACCEPTED_PASS inspection certificate is blocked immediately with HTTP 400 Bad Request (QUALITY_GATE_VIOLATION). No container truck can depart without digital QA clearance.

6.2 Customs-Compliant Export Commercial Invoicing (Mod 22)

Generates customs-compliant export commercial documentation conforming to international banking and Letter of Credit (LC) requirements:

Commercial Invoicing Parameters			Export Packing List & CBM Container Loading		
Invoice Number	EXP-INV-2026-0042		Total Garments Shipped	5,000 pieces	
Letter of Credit (LC)	LC-DB-2026-99014	(Deutsche Bank AG)	Export Cartons Packed	84 Cartons	(60 pcs/carton ratio)
Incoterms Rule	FOB Chittagong Port (BDCGP)		Automated Volume (CBM)	15.12 CBM	((L×W×H)/1,000,000)
Port of Discharge	Hamburg Port (DEHAM), Germany		Gross & Net Weights	Gross: 1,386.0 KG Net: 1,260.0 KG	
Invoice Commercial Value	\$42,500.00 USD (5,000 pcs @ \$8.50)		Allocated Container	40FT HQ (MSCU-991204-7)	
Payment Terms	Irrevocable At-Sight Letter of Credit		High-Security Bottle Seal	BD-SEAL-88912	(Customs Verified)

6.3 Plant Security Gate Pass & Live Highway Dispatch Tracking (Mod 23)

Security gate passes authorize the physical departure of container trucks from the factory compound:

GATE PASS #	TRANSPORT VEHICLE	DRIVER CREDENTIALS	CONTAINER SEAL #	DISPATCH STATUS	EXIT TIMESTAMP
GP-2026-0042	Covered Prime Mover DHAKA-METRO-DA-11-4455	Md. Anisur Rahman (+8801711-889900)	BD-SEAL-88912	GATE_OUT (IN-TRANSIT)	2026-06-25 17:15:00 BST

7. Phase 7: Factory Operations, Biometric Payroll & Compliance Suite

Phase 7 integrates plant operations: Human Resources, attendance, statutory payroll under the Bangladesh Labor Act 2006, machinery fleet maintenance, operational cost centers, and international social compliance audits.

7.1 HR Roster & Biometric Gate Attendance Ledger (Mod 24 & 25)

Manages the active workforce across Savar and Gazipur plants with biometric terminal integration (BI0-GATE-01) capturing shift clock-in/out and auto-computing daily overtime hours beyond the 8-hour general shift.

7.2 Bangladesh Labor Act 2006 Payroll Engine (Mod 25)

Statutory Standard: Enforced 2× Hourly Basic Overtime Rate

Under Section 108 of the Bangladesh Labor Act 2006, overtime wages for garment workers must equal exactly twice the ordinary hourly basic wage rate:

Hourly Basic Rate = Base Salary / 208 hours (26 days × 8 hours)

Overtime Hourly Rate = Hourly Basic Rate × 2.0

Monthly Net Pay = Base Salary + (Overtime Hours × Overtime Hourly Rate) + Attendance Bonus - Welfare Deductions

7.3 Production Machinery Fleet Asset Registry & Work Orders (Mod 26)

Machinery Asset Registry

Juki DDL-9000C	Direct Drive Lockstitch (Line 01) — Operational
Brother S-7200C	Electronic Lockstitch (Line 02) — Operational
Pegasus M900	5-Thread High-Speed Overlock (Line 03) — Breakdown
Kansai Special DX	Flatlock Interlock (Line 04) — Operational
Fleet Uptime Rate	75.0% Operational (3/4 active units)

Breakdown Work Orders & Spare Parts

Work Order #	MNT-2026-001	(Needle Bar Jam)
Severity Level	CRITICAL_STOPPAGE	
Assigned Mechanic	Shahadat Ali (Senior Plant Mechanic)	
Spare Parts Cost	৳1,250 BDT (Needle Bar Bushing replaced)	
Recorded Downtime	45 minutes (Machine restored to Operational)	

7.4 Operational Cost Centers & Realized Order Profitability (Mod 27)

Allocates factory overhead and calculates realized order net profitability:

PO-2026-001 Realized Financial Waterfall (H&M 5,000 pcs Polo Shirts @ \$8.50/pc FOB)

Export Commercial Revenue: \$42,500.00 (100.0%)

(-) Direct Fabric Cost: \$19,500.00 (45.9% | 100% Cotton Pique 3,925 KG)

(-) Trims & Packaging: \$3,250.00 (7.6% | Buttons, polybags, collar rib)

(-) Cost of Making (CM): \$6,750.00 (15.9% | Cutting, sewing, finishing labor)

(-) Factory Overhead: \$3,500.00 (8.2% | Utilities, commercial fees, machine amort)

(=) Total Cost of Production: \$33,000.00 (77.6% Total Cost Stack)

(=) Realized Net Profit: \$9,500.00 (22.4% Realized Net Margin → OPTIMAL_HIGH Rating)

7.5 International Social & Safety Compliance Audits (Mod 28)

Audit Number	Standard Framework	Auditing Agency	Audit Score	Overall Grade	CAPA Status
AUD-BSCI-2026-001	amfori BSCI Social Compliance	TUV Rheinland Bangladesh	94.5%	GREEN_COMPLIANT	Zero Critical Findings Certified Closed
AUD-RSC-2026-002	RSC / Accord Building & Fire Safety	RMG Sustainability Council	88.0%	YELLOW_MINOR_CAPA	Emergency exit illumination module fitted Verified

8. Phase 8: System Intelligence, Order Traceability 360° & Executive Plant BI

Phase 8 serves as the executive intelligence engine of GarmentERP BD, consolidating operational data from all 7 prior phases into real-time digital passport tracking, plant OEE analytics, and automated telemetry alerts.

8.1 Order Traceability 360° Digital Passport (Mod 29)

Every Buyer PO possesses an auditable 10-milestone digital passport verifying quality gates, operator signatures, and process milestones:

#	MILESTONE CODE	STAGE NAME	DEPARTMENT	GATE STATUS	KEY VERIFICATION OUTPUT
01	BUYER_PO_CONTRACT	Buyer PO Registration	Merchandising	PASSED	10,000 pcs confirmed @ \$8.00 (\$80,000 Revenue)
02	TECHPACK_COSTING	Tech Pack & BOM Approval	Product Dev	PASSED	Tech Pack v1.2 approved; Target margin: 22.4%
03	RAW_MATERIAL_SCM	Material SCM & GRN	Supply Chain	PASSED	3,925 kg Cotton Pique received under GRN-2026-001
04	FABRIC_4POINT_QC	Fabric 4-Point Inspection	Quality Assurance	PASSED	Roll score: 2.0 pts/100 sq yds (FIRST_QUALITY_PASS)
05	CUTTING_BUNDLING	24h Relaxation & QR Bundles	Cutting Section	PASSED	24h relaxation verified; 84 QR bundles generated
06	SEWING_PRODUCTION	Sewing Assembly (Line 01)	Sewing Dept	PASSED	5,000 pcs assembled at 91.4% line efficiency
07	INLINE_QA_REWORK	Inline QC & Rework Loop	Quality Assurance	PASSED	1.6% DHU; 4 defects repaired & RE_INSPECTED_PASS
08	FINISHING_PACKING	Steam Press & Carton Packing	Finishing & Pack	PASSED	Metal detector pass; 84 export cartons (15.12 CBM)
09	AQL_PRE_SHIPMENT	ISO 2859-1 AQL 2.5 Audit	Quality Assurance	PASSED	315 pcs sampled; 0 Critical; ACCEPTED_PASS Certificate
10	EXPORT_DISPATCH	Invoice & Gate Out Dispatch	Commercial Logistics	PASSED	Invoice EXP-INV-2026-0042; Gate Pass GP-0042; Gate Out

8.2 Executive Plant BI & Overall Equipment Effectiveness (OEE) (Mod 30)

Plant Floor OEE Calculation Availability Rate: 94.2% (Operational uptime) Performance Rate: 91.8% (Actual vs standard target output) Quality Rate: 98.4% (First-time-right yield) Overall OEE: 85.1% (World-Class Target > 80%)	Top 5 Sewing Defect Pareto Analysis <table><tr><td>1. Skipped Stitch (Collar/Placket)</td><td>34.6% (18 pcs)</td></tr><tr><td>2. Seam Puckering (Side Seam)</td><td>26.9% (14 pcs)</td></tr><tr><td>3. Broken Needle Fragment</td><td>15.4% (8 pcs)</td></tr><tr><td>4. Uneven Bottom Hem</td><td>13.5% (7 pcs)</td></tr><tr><td>5. Oil Stains / Lubricant</td><td>9.6% (5 pcs)</td></tr></table>	1. Skipped Stitch (Collar/Placket)	34.6% (18 pcs)	2. Seam Puckering (Side Seam)	26.9% (14 pcs)	3. Broken Needle Fragment	15.4% (8 pcs)	4. Uneven Bottom Hem	13.5% (7 pcs)	5. Oil Stains / Lubricant	9.6% (5 pcs)
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5. Oil Stains / Lubricant	9.6% (5 pcs)										

8.3 Multi-Factory Comparative Benchmarking: Savar vs. Gazipur

MANUFACTURING UNIT	LOCATION	ACTIVE LINES	DAILY CAPACITY	ACTUAL OUTPUT	LINE EFFICIENCY	AVERAGE DHU %	WORKFORCE
Apex Garments — Dhaka Unit (Savar)	Savar, Dhaka	4 Lines	4,000 pcs	3,650 pcs	91.25%	1.2%	120 Operators
Apex Garments — Gazipur Complex	Kashimpur, Gazipur	4 Lines	3,800 pcs	3,380 pcs	88.90%	1.4%	110 Operators

9. Complete SQA Automated Test Suite & Architectural Signoff

GarmentERP BD enforces rigorous Software Quality Assurance (SQA) across all 8 development phases, with automated integration test suites guaranteeing zero-defect deployments.

9.1 Automated Multi-Module Continuous Integration Matrix (96 Tests)

Development Phase	Functional Scope	Automated Tests	Test Pass Rate	Regression Suite File
Phase 1: Foundation & RBAC	Auth, JWT, 14 RBAC Roles, Multi-Factory Tree, Audit Trail	10 Tests	10 / 10 PASS (100%)	dist/tests/auth.test.js
Phase 2: Merchandising	Buyer Profiles, Tech Pack v1-v3, Pre-Costing BOM, Buyer POs	10 Tests	10 / 10 PASS (100%)	dist/tests/merchandising.test.js
Phase 3: SCM & Inventory	Supplier Scorecard, PR, SPO, GRN, Stock Balance Lock	12 Tests	12 / 12 PASS (100%)	dist/tests/supplychain.test.js
Phase 4: Garments Production	24h Relaxation, Spreading, QR Bundles, Lines 01-04, Packing	12 Tests	12 / 12 PASS (100%)	dist/tests/production.test.js
Phase 5: Core QA/QC Suite	Fabric 4-Point, Inline DHU, Rework Orders, CAPA, AQL 2.5	12 Tests	12 / 12 PASS (100%)	dist/tests/qa.test.js
Phase 6: Shipment & Logistics	AQL Quality Gate Lock, Invoicing, Packing List (CBM), Gate Pass	12 Tests	12 / 12 PASS (100%)	dist/tests/shipment.test.js
Phase 7: Factory Operations	HR, Biometric Punch, 2x OT Payroll, Maintenance, Audits	14 Tests	14 / 14 PASS (100%)	dist/tests/operations.test.js
Phase 8: System Intelligence	Order 360° Digital Passport, Executive BI, OEE, Alerts Engine	14 Tests	14 / 14 PASS (100%)	dist/tests/analytics.test.js
TOTAL PLATFORM COVERAGE	Full RMG Manufacturing ERP & Quality System	96 Tests	96 / 96 PASS (100%)	scripts/test-all.js

9.2 Postman SQA Test Collections (Ready for CI/CD Newman Execution)

Core Manufacturing Collections	Quality & Intelligence Collections
Phase 1 GarmentERP_Foundation_Phase1.postman_collection.json	Phase 5 GarmentERP_QA_QC_Phase5.postman_collection.json
PhaseGarmentERP_Merchandising_Phase2.postman_collection.json 2	Phase 6 GarmentERP_Shipment_Phase6.postman_collection.json
Phase 3 GarmentERP_SupplyChain_Phase3.postman_collection.json	Phase 7 GarmentERP_Operations_Phase7.postman_collection.json
Phase 4 GarmentERP_Production_Phase4.postman_collection.json	PhaseGarmentERP_Intelligence_Phase8.postman_collection.json 8

9.3 Architectural Signoff & Sovereignty Declaration

Enterprise Ready: Complete Bangladesh RMG ERP & SQA Suite

This specification certifies that **GarmentERP BD** is fully designed, engineered, automated, and verified across all 8 development phases. From initial Buyer Order intake to Cutting, Sewing, Quality Assurance, Packing, Commercial Invoicing, Gate Pass Dispatch, Statutory Payroll, and Executive BI Traceability 360°, the platform provides a unified, single source of truth for modern garment manufacturing in Bangladesh.